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Requirements Table

Problem Statement #35 — Customizable Subscription Box Scheduler

Exactly five Functional Requirements (FR-001–FR-005) and two Non-Functional Requirements (NFR-001, NFR-002). FR-001 and NFR-001 were provided in the problem statement; the remaining requirements were elicited from the scenario.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow subscribers to swap monthly box items or pause deliveries up to 48 hours prior to the monthly billing renewal date.	High	Pass: Selection updated and fulfillment manifest refreshed. Fail: Customization locked out prematurely.	Core value proposition — subscribers must retain control over their box before it enters fulfillment. (Given)
FR-002	Functional	The system shall allow subscribers to set and update preference tags (e.g., dietary, genre, roast type) used to auto-curate their monthly box selections.	High	Pass: Saved tags are applied to the next auto-curated box and reflected in the subscriber's profile within 5 seconds. Fail: Tag changes are not reflected in the next curation cycle.	Personalization based on preference tags is the primary differentiator described in the problem statement.
FR-003	Functional	The system shall allow subscribers to skip a single upcoming billing cycle without cancelling their subscription, provided the request is submitted at least 48 hours before the renewal date.	Medium	Pass: No charge is generated and no box is scheduled for the skipped cycle; subscription auto-resumes the following month. Fail: Subscriber is charged or a box is shipped for a skipped cycle.	Reduces involuntary churn by giving subscribers flexibility instead of forcing outright cancellation.
FR-004	Functional	The system shall automatically charge the subscriber's saved payment method on the billing renewal date and update the subscription status accordingly.	High	Pass: Successful charge updates status to "Active – Paid" and triggers manifest inclusion within 1 minute. Fail: Subscription proceeds to fulfillment without a confirmed charge.	Automated billing is required for the subscription model to function without manual intervention.
FR-005	Functional	The system shall allow the Fulfillment Lead to generate and export a fulfillment manifest (box contents and shipping labels) for all confirmed subscriber boxes ahead of the shipping date.	High	Pass: Exported manifest lists every confirmed box with correct address and contents, matching the confirmed-orders count. Fail: Manifest omits or duplicates any confirmed box.	Fulfillment operations depend on an accurate, exportable manifest to pack and ship on schedule.
NFR-001	Nonfunctional	The fulfillment manifest generator must export shipping labels for 10,000 monthly boxes in under 1 minute.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Performance & Security — the fulfillment window is fixed, so manifest generation must scale with subscriber growth. (Given)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-002	Nonfunctional	The system shall encrypt all stored subscriber payment information and process transactions in compliance with PCI-DSS standards.	High	Pass: Independent security audit confirms payment data is encrypted at rest and in transit, and PCI-DSS controls pass validation. Fail: Any payment field is stored or transmitted unencrypted.	Security — recurring billing requires storing payment credentials, making data protection a compliance necessity, not an option.

Actors & Use Cases

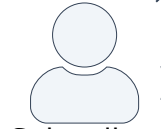
Actors

Actor	Role	Description
Subscriber	Primary	The end customer who owns a subscription and customizes, pauses, skips, or pays for their monthly box.
Fulfillment Lead	Secondary (Human)	Warehouse/operations staff who generate and export the fulfillment manifest for shipping.
Payment Gateway	Secondary (System)	External payment processing system that validates and authorizes subscriber charges.

Use Cases

Use Case	Name	Related Requirement	Notes
UC-01	Set Preference Tags	FR-002	Initiated by Subscriber.
UC-02	Swap or Pause Monthly Delivery	FR-001	Initiated by Subscriber; see use-case-flow.docx for the full spec.
UC-03	Skip Billing Cycle	FR-003	Initiated by Subscriber.
UC-04	Process Monthly Payment	FR-004	Initiated by Subscriber (trigger) / system-scheduled; «include» UC-06; extended by UC-07.
UC-05	Generate Fulfillment Manifest	FR-005 & NFR-001	Initiated by Fulfillment Lead.
UC-06	Validate Payment Details	NFR-002	Included by UC-04 (always performed); interacts with Payment Gateway.
UC-07	Apply Promotional Discount	— (supporting)	Extends UC-04 only when a valid promo code is present (optional).

Customizable Subscription Box Scheduler



Subscriber

UC-01
Set Preference Tags

UC-02
Swap or Pause
Monthly Delivery

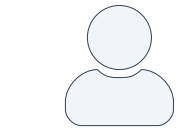
UC-03
Skip Billing Cycle

UC-07
Apply Promotional
Discount

UC-05
Generate Fulfillment
Manifest

UC-04
Process Monthly
Payment

UC-06
Validate Payment
Details



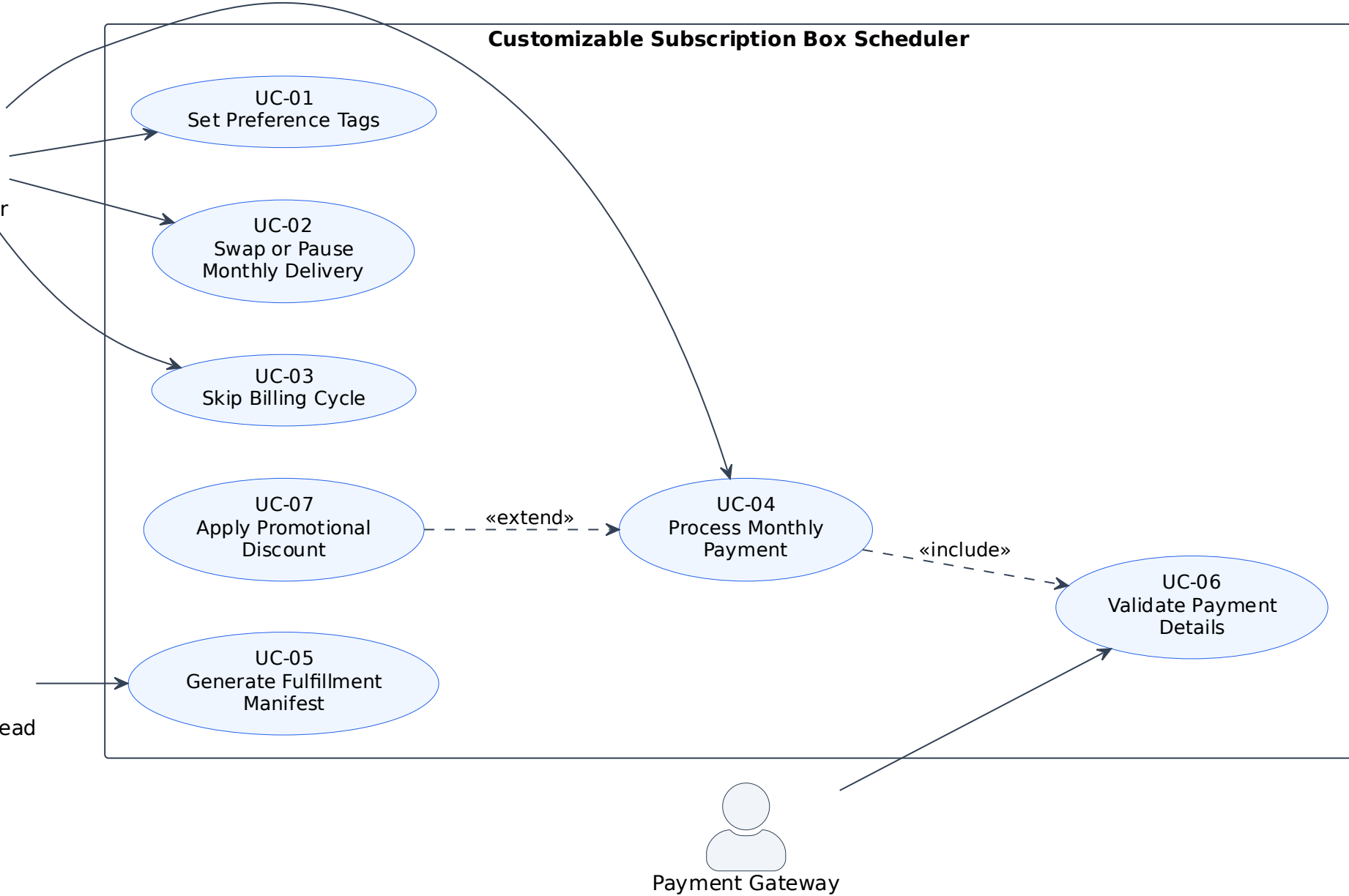
Fulfillment Lead



Payment Gateway

«extend»

«include»



Use-Case Flow Specification

UC-02 – Swap or Pause Monthly Delivery (implements FR-001)

Primary Actor: Subscriber

Secondary Actor: — (none required for this flow)

Preconditions

- The Subscriber is authenticated and logged into the subscription portal.
- The Subscriber has an active subscription with an upcoming billing renewal date.
- The current date/time is before the 48-hour customization cutoff for the upcoming cycle.

Postconditions (on success)

- The subscriber's box selection is updated, or the upcoming delivery is marked as paused, in the subscriber's profile.
- The fulfillment manifest for the upcoming cycle is refreshed to reflect the change.
- The subscriber receives an on-screen and emailed confirmation of the change.

Main Success Scenario

1. Subscriber logs into the subscription portal and navigates to “My Upcoming Box.”
2. System displays the current box selection, scheduled delivery date, and a countdown to the 48-hour customization cutoff.
3. Subscriber chooses to either (a) swap one or more box items using preference-tag-filtered alternatives, or (b) pause the upcoming delivery.
4. System validates that the current time is before the 48-hour cutoff.
5. Subscriber confirms the requested change.
6. System updates the subscriber's box selection (or pause status) in the database.
7. System refreshes the fulfillment manifest so it reflects the updated selection.
8. System displays a confirmation message and sends a confirmation email/notification.
9. Use case ends successfully.

Alternate Flow — 4a. Customization Window Closed

- a. System detects that the request was submitted after the 48-hour cutoff.
- b. System displays a “Customization locked” message explaining that the box has already entered fulfillment processing.
- c. System offers the Subscriber the option to apply the requested change to the following month's cycle instead.
- d. If the Subscriber agrees, the system schedules the change for the next billing cycle and confirms it; otherwise the use case ends with no changes made to the current cycle.